

SECURE YOUR LINUX BOX WITH EFFECTIVE USER AND GROUP MANAGEMENT

Security is a paramount concern today with identity theft becoming an everyday occurrence. Linux security is built around privileges given to users and groups. This article introduces the reader to the basics of user and group management in Linux.



Linux is basically a kernel on which various distributions of operating systems have been developed, namely Red Hat, Fedora, Ubuntu, SUSE, etc. Since its inception, Linux has become the most dominant technology used to manage servers worldwide. In recent years, its popularity has extended to normal users also, rather than just being the administrators' preferred choice. No doubt, one of the compelling attractions of Linux is that it is open source but another reason for choosing Linux for middle to high-end machines is that Linux is safe and secure.

A common myth is that Linux is built up on only plain text files, and that it is 100 per cent virus-free. However, this is not the truth. Hackers and crackers have always tried to inject threats into the Linux environment, and initially, they were successful to a certain extent. The reason for Linux's security is that all its processes run strictly under the privileges allocated to various users by the systems administrator. Privileges, if implemented fairly well, make an unbreakable security layer around the Linux engine and prevent it from being attacked. In Linux, users and group members access a file system's contents (files and directories) based on the privileges assigned to them in the

form of permissions. This is further enhanced by Access Control Lists, Sticky Bits and Security Enhanced Linux (SELinux).

Users and groups

A **user** is a person who is authorised to use the Linux interface under privileges issued by the systems administrator. Every person gets a username and password as credentials, which are to be passed during the login process for receiving an interactive shell. When we add a new user, several details are automatically configured by Linux. However, these can be customised as we will see later. There are five kinds of users:

- **System user** – works for system applications and has a non-interactive shell. These users need not log in; rather, they become active as their corresponding system services start.
- **Super user** – is the one who has full control over the Linux file system and is empowered with unlimited permissions. Root is the default super user.
- **Owner user** – is the one who is the creator or owner of the content and uses the allotted permissions.